

Installation Guide: Distributed Math Wizard

Prerequisites

- Install Docker and Docker Compose on your host machine.
- Ensure you have the necessary identity policy credentials (e.g., University Email) if deploying in Public Mode.

Step 1: Prepare the Environment

- Extract the submitted `Code` directory from the project zip file.
- Open a terminal or command prompt.
- Navigate to the root folder where the `docker-compose.yml` file is located.

```
(base) polloz@DESKTOP-NCKF390 ~/Dev/TheDistributedMathWizard (Rayhan) $ ls
Dockerfile  app                jtestresults.png  requirements.txt  test
README.md   docker-compose.yml nginx              secrets           wizards_test.jmx
(base) polloz@DESKTOP-NCKF390 ~/Dev/TheDistributedMathWizard (Rayhan) $
```

Step 2: Launch Local Development Mode

- Use this mode for testing without internet exposure.
- Execute the command: `docker compose up --build`
- The system will build and start the Nginx gateway (Port 80), 3 API/Worker containers, and the Valkey state database.

```
(base) polloz@DESKTOP-NCKF390 ~/Dev/TheDistributedMathWizard (Rayhan) $ docker compose up --build
WARN[0000] /home/polloz/Dev/TheDistributedMathWizard/docker-compose.yml: the attribute 'version' is obsolete, it will be
ignored, please remove it to avoid potential confusion
[+] Building 3.5s (5/10)
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 1.08kB 0.0s
=> [dlq_wizard internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 278B 0.0s
=> [math_wizard internal] load metadata for docker.io/library/python:3.11 0.4s
=> [math_wizard internal] load docker image 0.0s
=> => transferring context: 237B 0.0s
=> [math_wizard 1/5] FROM docker.io/library/python:3.11@sha256:5062f6c22a2c3b8b3717c642af3852310d9a81c3c2c0fc724 2.0s
=> => resolve docker.io/library/python:3.11@sha256:5062f6c22a2c3b8b3717c642af3852310d9a81c3c2c0fc724 0.0s
=> => sha256:5062f6c22a2c3b8b3717c642af3852310d9a81c3c2c0fc724 10.52kB / 10.52kB 0.0s
=> => sha256:2b60ad636ac665a0ba3bf995c6a705780e214fd468d0f8d4fad4f9ffcee13a 2.32kB / 2.32kB 0.0s
=> => sha256:20d12c8f3f3fbb5bd2b8369adf3213e09d6b33f975462543ddfece85b2fe85e5 10.49MB / 67.79MB 2.0s
=> => sha256:1683e38056fb8e0eca412944f7b7e7573553923ccc26889770bb413d317b8 6.35kB / 6.35kB 0.0s
=> => sha256:3b32e3bb7338c216b077e95920ae53332d81bd57f4a7393bc324432d5a3f88a2 31.46MB / 49.30MB 2.0s
=> => sha256:5467f93954cfe1451f4333422086d00bd066cf33f42112b531804ffe06f0a929 23.07MB / 25.62MB 2.0s
=> [math_wizard internal] load build definition 0.0s
=> => transferring context: 167.43kB 0.0s
```

Step 3: Launch Public Production Mode (Alternative)

- Use this mode to expose the app securely via Cloudflare Tunnel.
- Execute the command: `docker compose --profile public up --build`
- This starts all core services plus the `cloudflared` sidecar for Zero Trust Access (HTTPS Port 443).

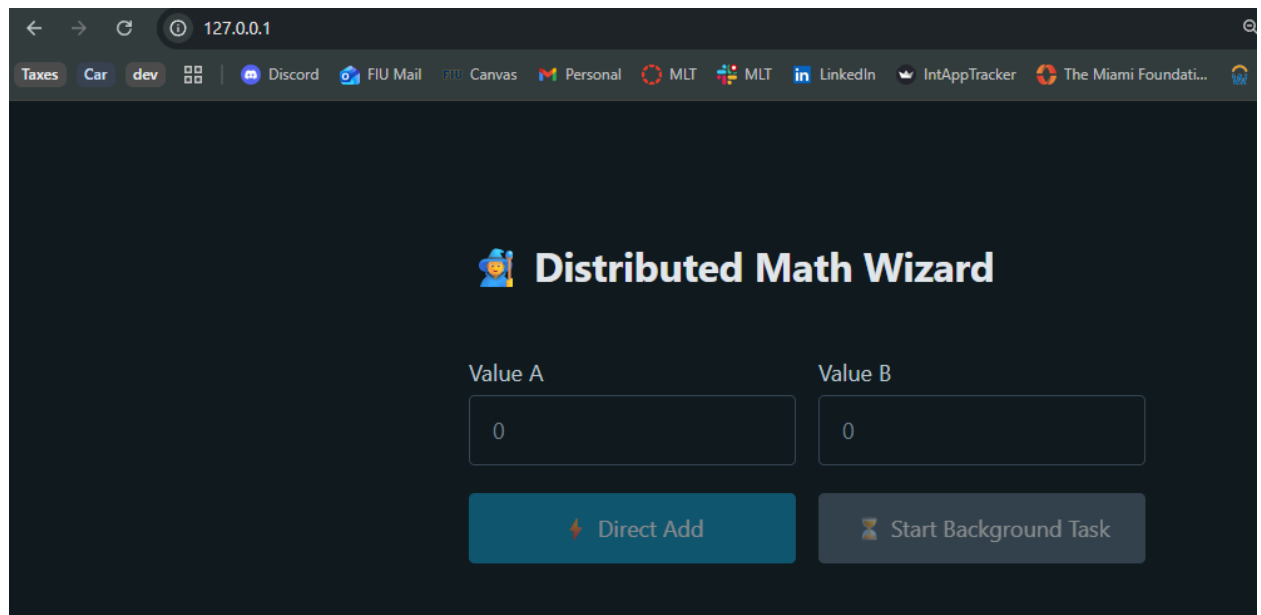
```

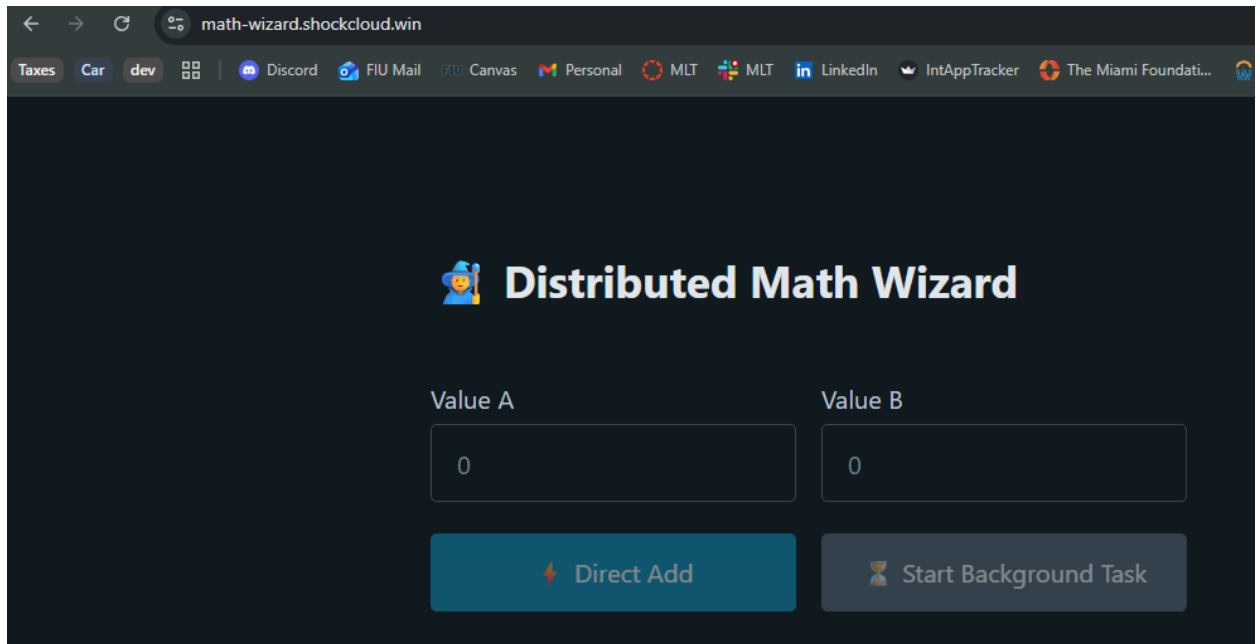
nginx_proxy | /docker-entrypoint.sh: Configuration complete; ready for start up
tunnel-1 | 2026-04-28T00:00:24Z INF Starting tunnel tunnelID=8d28e1d4-a279-48d5-9571-0029503466f8
tunnel-1 | 2026-04-28T00:00:24Z INF Version 2025.2.1 (Checksum 573246eb7783bbeabb1cb40b0ec86817876eaf
dd759d5a75ee2)
tunnel-1 | 2026-04-28T00:00:24Z INF GOOS: linux, GOVersion: go1.22.5-devel-cf, GoArch: amd64
tunnel-1 | 2026-04-28T00:00:24Z INF Settings: map[no-autoupdate:true token:****]
tunnel-1 | 2026-04-28T00:00:24Z INF Environmental variables map[TUNNEL_TOKEN:****]
tunnel-1 | 2026-04-28T00:00:24Z INF Generated Connector ID: 7b1bfeca-da3d-4517-b953-02771a1bc032
tunnel-1 | 2026-04-28T00:00:24Z INF Initial protocol quic
tunnel-1 | 2026-04-28T00:00:24Z INF ICMP proxy will use 172.18.0.8 as source for IPv4
tunnel-1 | 2026-04-28T00:00:24Z INF ICMP proxy will use ::1 in zone lo as source for IPv6
tunnel-1 | 2026-04-28T00:00:24Z INF ICMP proxy will use 172.18.0.8 as source for IPv4
tunnel-1 | 2026-04-28T00:00:24Z INF ICMP proxy will use ::1 in zone lo as source for IPv6
tunnel-1 | 2026-04-28T00:00:24Z INF Starting metrics server on [::]:20241/metrics
tunnel-1 | 2026-04-28T00:00:24Z INF Using [CurveID(4588) CurveID(25497) CurveP256] as curve preferenc
0 event=0 ip=198.41.200.23
tunnel-1 | 2026/04/28 00:00:24 failed to sufficiently increase receive buffer size (was: 208 kiB, war
, got: 416 kiB). See https://github.com/quic-go/quic-go/wiki/UDP-Buffer-Sizes for details.
tunnel-1 | 2026-04-28T00:00:24Z INF Registered tunnel connection connIndex=0 connection=d79de520-aad6
6b1389951d event=0 ip=198.41.200.23 location=mia04 protocol=quic
tunnel-1 | 2026-04-28T00:00:24Z INF Using [CurveID(4588) CurveID(25497) CurveP256] as curve preferenc
1 event=0 ip=198.41.192.57
tunnel-1 | 2026-04-28T00:00:24Z INF Updated to new configuration config="{\"ingress\":{\"hostname\":
.shockcloud.win\", \"originRequest\":{\"originServerName\": \"math-wizard.shockcloud.win\"}, \"service\": \"htt
oxy:443\"}, {\"service\": \"http_status:404\"}}, {\"warp-routing\":{\"enabled\": false}}\" version=5
tunnel-1 | 2026-04-28T00:00:24Z INF Registered tunnel connection connIndex=1 connection=93d6f928-2dc2
d8672ce07c event=0 ip=198.41.192.57 location=mia02 protocol=quic

```

Step 4: Verification

- For Local Mode, open a web browser and navigate to <http://localhost>.
- For Public Mode, navigate to <https://math-wizard.shockcloud.win> and authenticate if prompted by the Zero Trust policy.
- Ensure the main UI dashboard loads successfully.





Step 5: Shutting Down

- To stop the cluster, press `Ctrl+C` in the terminal where the containers are running.
- To completely remove the containers and network, run: `docker compose down`